

HE-IFD: Privacy-Preserving One-Shot Federated Fine-Tuning of Pretrained Models under Homomorphic Encryption

Halil İbrahim Kanpak , Sinem Sav , Alptekin Küpçü 

Abstract—Federated learning lets parties train a shared model without pooling their data, but its one-shot variant, which communicates in a single round, faces an unresolved privacy dilemma: sending a client’s contribution in plaintext exposes it, while protecting it with differential privacy injects noise that degrades the released model and offers only a statistical, not a cryptographic, guarantee, and homomorphic encryption has so far been confined to multi-round encrypted training or to secure-aggregation primitives. We present HE-IFD, a one-shot federated fine-tuning whose contributions are protected by homomorphic encryption. Each client fine-tunes a small adapter on a frozen pretrained backbone for a bounded number of steps, and the server combines the resulting parameter displacements with a single sample-weighted linear operation under multiparty CKKS, at multiplicative depth one. The key to making a one-shot linear aggregation work under heterogeneous data is a shared frame, which a frozen pretrained backbone supplies for free: every client fine-tunes from the same public initialization, so there is no alignment phase and nothing data-derived leaves a client before encryption. Across vision and language tasks on frozen pretrained backbones, the method produces a common model stronger than any client’s own and close to a centralized one, while giving each client coverage of classes it never observed locally. A real multiparty CKKS implementation reproduces the plaintext aggregate within the approximation bounds of CKKS at a few megabytes per round.

Index Terms—Federated learning, one-shot federated learning, federated fine-tuning, homomorphic encryption, multiparty computation, privacy-preserving machine learning.

I. INTRODUCTION

Many organizations would benefit from training a shared model on their combined data but cannot pool it, whether for legal, competitive, or privacy reasons. Federated learning addresses this by exchanging model updates instead of data, but in its usual form it does so over many rounds of communication, and each round discloses a fresh update computed on private data. One-shot federated learning removes the first cost by collapsing the entire interaction into a single upload and download. It does not, by itself, remove the second. A client’s

single contribution still encodes its private data, and the field’s answers to this have been unsatisfying at the extremes: sending the contribution in plaintext exposes it, while protecting it with differential privacy injects noise into the released model and provides only a statistical, not a cryptographic, guarantee. Homomorphic encryption promises a way out by operating on contributions while they stay encrypted, but it has been applied to federated learning either as multi-round encrypted training, whose cost is measured in hours and gigabytes, or as a secure-aggregation primitive embedded in an iterative protocol. No existing method delivers a one-shot federated distillation whose contributions are protected cryptographically rather than by accuracy-degrading noise.

We present **HE-IFD**. Each client distills its own locally trained teacher into a small shared classifier head, and the server combines the clients’ contributions with a single linear operation, $\theta^* = \theta_0 + \sum_j w_j \Delta_j$, evaluated under multiparty CKKS. The server holds the shared initialization θ_0 as a public constant and never modifies it during the protocol: it only adds the aggregated displacement to this preserved model, so no training takes place on the server side. Because the combination is linear, it maps onto the operations a homomorphic scheme evaluates at multiplicative depth one, without bootstrapping, and moves a few megabytes per round rather than hundreds of gigabytes. The privacy of the contributions is therefore cryptographic, and because the feature extractor is frozen and never enters the encrypted combination, the same protocol and the same cost carry from a small from-scratch network to a frozen pretrained backbone.

The difficulty a one-shot linear aggregation must overcome is that independently trained models do not average well under heterogeneous data: starting from different initializations and fitting different local distributions, their updates point in conflicting directions and cancel when summed. Fine-tuning a frozen pretrained backbone removes this difficulty without any alignment phase. The fixed backbone places every client in a common representation, and every client starts its small trainable unit from the same public initialization, so their bounded updates are expressed in one frame and reinforce rather than cancel. Where prior one-shot federated learning must first agree on a starting point by exchanging per-class summaries — an extra round that exposes data-derived quantities and, when protected, spends a differential-

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privacy budget — a pretrained backbone supplies the shared frame for free, and nothing data-derived leaves a client before encryption.

We show across vision and language tasks, and from small networks to frozen vision transformers and sentence encoders, that the protocol produces a common model stronger than any client’s own and close to a centralized model trained on the pooled data, while every client gains the ability to classify categories it never observed locally. We confirm in a real multiparty CKKS implementation that the encrypted aggregation reproduces the plaintext result within the approximation bounds of CKKS, at a cost of a few megabytes per round, and we trace the privacy–utility frontier of the basin summaries to show that their accuracy cost is negligible across the single-digit privacy budgets typical of differentially private learning.

Our contributions are as follows.

- We introduce, to our knowledge, the first one-shot federated fine-tuning whose contributions are protected by homomorphic encryption, with a single server operation of multiplicative depth one under multiparty CKKS (section IV).
- We show that fine-tuning a frozen pretrained backbone supplies the shared frame a one-shot linear aggregation needs at no cost, eliminating the data-dependent alignment phase that prior one-shot federated learning requires — and with it the leakage and the differential-privacy budget that phase entails (sections IV and V-B).
- We give a privacy design in which the contributions are protected by cryptography alone, with no perturbation and no differential privacy, so the only quantity exposed in the clear is the final shared model, whose residual leakage we measure directly with membership inference (section V-I).
- We demonstrate that the method yields a common, superior model across frozen pretrained backbones in both modalities, and validate the encrypted aggregation in a real multiparty CKKS implementation at a cost of a few megabytes per round (sections V-B and V-H).

II. RELATED WORK

The setting we study sits at the meeting point of three lines of work that have largely developed in isolation: how federated learning communicates, how it is made private, and how it has been compressed into a single round. Tracing how each arrived at its present state makes clear both what is established and where the gap we address lies.

Federated learning began as an iterative procedure. In the original formulation a server repeatedly broadcasts a global model, clients improve it on their local data, and the server averages the returned updates [1]. This works well but inherits two persistent costs from its iterative nature. The first is communication: reaching a good model can take tens or hundreds of rounds, each moving a full model in both directions. The second is exposure: every round reveals a fresh update computed on private data, and a long sequence of such updates is a rich target for inference. Much of the subsequent literature can be read as attempts to control one of these two costs without worsening the other.

The exposure problem has been met with two broad tools, distinguished by whether they alter the model that is ultimately released. Secure aggregation lets a server learn only the sum of the clients’ updates and nothing about any individual one, using masking protocols that cancel in the aggregate [2]; it is lossless, in that the aggregate is exact, but it protects only the per-round sum and still operates within the iterative protocol. Differential privacy instead perturbs the updates themselves, most commonly through differentially private stochastic gradient descent [3]; the guarantee is strong and composable, but because the noise enters the released model it trades privacy directly against accuracy. This distinction, between privacy that costs accuracy and privacy that does not, is the axis along which our own privacy design is best understood, and it recurs throughout what follows.

Homomorphic encryption offers a third route, protecting the updates by computing on them while they remain encrypted. In its most ambitious form it has been used to train a neural network entirely under encryption across many parties. POSEIDON, for instance, runs encrypted stochastic gradient descent under multiparty CKKS, with the nonlinear activations replaced by polynomial approximations so that they can be evaluated homomorphically [4]. The privacy adds no accuracy cost and the result is strong, but the approach remains fundamentally iterative and pays for the encrypted nonlinearities with a deep multiplicative circuit, repeated bootstrapping, and a wall-clock measured in hours. A lighter line keeps the encryption but narrows its job to aggregation alone: schemes that combine multi-key CKKS with masking reduce a round of secure summation to a single non-interactive client-to-server transmission [5]. These are efficient and genuinely cryptographic, but they encrypt the sum of model updates within an iterative training protocol rather than a one-shot contribution, and they perform aggregation rather than knowledge transfer. On the encryption axis these works are the closest to ours, and the contrast is precise: they are either multi-round or are secure-aggregation primitives, whereas we encrypt a single distillation displacement.

Homomorphic encryption has also been applied to federated learning purely for aggregation, encrypting the per-round updates so the server can sum them without seeing any in the clear; BatchCrypt, FedML-HE, and FedSHE develop this for cross-silo training with various packing and quantization optimisations [6], [7], [8]. These remain multi-round and, like secure aggregation, protect the sum rather than transferring knowledge. A recurring obstacle across all encrypted-training work is the nonlinearity: activations must be replaced by polynomial approximations to be evaluated homomorphically, which deepens the multiplicative circuit and is delicate to train, as a line of work on polynomial activations and encryption-friendly networks documents [9], [10], [11], [12], [13]. Our design sidesteps this entirely: the only encrypted operation is linear, so no activation is ever evaluated under encryption and the circuit stays at depth one.

The communication problem has driven the parallel development of one-shot federated learning, which collapses the entire exchange into a single upload and download [14], [15]. The earliest approaches transferred knowledge rather

than weights, having clients exchange predictions on a shared public set so that heterogeneous models could teach one another [16], [17]; in practice these remained iterative, repeating the exchange to converge. Ensemble distillation made the distillation step explicit by training a server model against the clients' aggregated predictions, though in its standard configuration it too runs for many rounds [18]. The decisive step to a true single round came with data-free one-shot methods, which avoid any shared data by having the server distil the clients' models through a generator or through synthesised inputs, as in DENSE and its successors [19], [20], and with fusion-based methods that stitch client models together with lightweight learned adapters [21] or aggregate them through a layer-wise posterior [22], [23]. These methods are the closest to ours on the one-shot and distillation axes, and they establish that a single round can produce a usable joint model. What they do not provide is any cryptographic protection: they operate in plaintext, so the contributions each client sends are exposed.

Closing that last gap, a handful of works add privacy to one-shot federated learning, and they do so with differential privacy. FedAUXfdp distils through a pretrained feature extractor on auxiliary public data and releases the client contributions under a differentially private mechanism [24]; related approaches protect a diffusion-generated [25] or a teacher-ensemble [26] surrogate, or add noise-free differential privacy to the distillation itself [27], under the same kind of guarantee. These are the natural quantitative peers for our method, since they share its one-shot, distillation-based structure, but their privacy is lossy in exactly the sense above: the noise needed for a meaningful budget is added to the artifact that is released, and accuracy falls as the budget tightens.

The reason privacy is the binding concern here is that the artifacts federated protocols exchange leak. Sharing gradients or model updates permits reconstruction of training inputs [28], [29], [30] and supports membership and property inference [31], [32], [33], [34], [35], [36], and even released confidences enable model inversion [37], [38]. Federated distillation, which shares predictions or a public-data signal instead of gradients, was hoped to sidestep this, but a growing body of attacks shows it leaks as well: paired-logit inversion recovers images from shared logits [39], and membership can be inferred from distillation outputs and from public-dataset usage [40], [41], [42], [43], [44]. This is why we protect the contributions cryptographically rather than by perturbation, and why we measure in section V-I what the released model still reveals.

Seen together, the two strongest lines never meet. The cryptographic work that would protect contributions without loss is either multi-round encrypted training or a secure-aggregation primitive, and the one-shot distillation work that communicates in a single round is either plaintext or made private only by lossy perturbation. The method we present occupies the empty intersection: a one-shot federated distillation whose single encrypted operation is a linear aggregation under multiparty homomorphic encryption, giving the accuracy-preserving privacy of the cryptographic line at the communication budget of the one-shot line.

TABLE I
NOTATION.

Symbol	Meaning
N	number of participating clients
$\mathcal{D}_j, n_j$	client j 's local dataset and its size $n_j = \mathcal{D}_j $
C	number of classes
α	Dirichlet concentration (label heterogeneity)
φ	frozen public backbone (never trained or encrypted)
ψ	parameters of the trainable unit (adapter + head)
f_θ	model on the frozen features, parameters θ
θ_0	shared public initialization of the trainable unit
K	local fine-tuning steps
Δ_j	displacement $\theta_j^{(K)} - \theta_0$
w_j	sample weight $n_j / \sum_i n_i$
θ^*	aggregated global model $\theta_0 + \sum_j w_j \Delta_j$

III. PRELIMINARIES AND THREAT MODEL

A. Notation

Table I collects the symbols used throughout. Client indices are $j \in \{1, \dots, N\}$.

B. Multiparty Homomorphic Encryption

We build on the CKKS scheme [45], a ring-learning-with-errors construction [46] that performs approximate arithmetic on vectors of real numbers under encryption. CKKS packs many real slots into one ciphertext and supports two homomorphic operations of interest here: addition of two ciphertexts, and multiplication of a ciphertext by a plaintext. It is a levelled scheme, meaning each multiplication consumes one level of a finite budget; our protocol uses only additions and multiplications by public plaintext scalars, so it consumes a single level and never requires the expensive bootstrapping step [47] that refreshes a depleted budget.

A single-key scheme would require one party to hold the secret key and therefore be trusted with decryption. We instead use the multiparty variant of CKKS [48], in which the secret key is shared among the clients through a distributed key-generation protocol that produces a collective public key without ever materialising the secret in one place. Any party, including the server, may compute on ciphertexts under the collective public key, but decryption requires the cooperation of a threshold of clients, carried out as a collective key-switching protocol. No single party, and in particular not the server, can decrypt on its own. This is exactly the property our aggregation needs: the server performs the homomorphic computation on encrypted contributions, yet only the clients, acting together, can reveal the result.

C. Threat Model

The participants are N clients and a single aggregation server. The server is honest-but-curious: it follows the protocol but may try to infer private information from anything it observes. During aggregation it sees only ciphertexts under the collective key, and as established in section III-B it cannot decrypt them. A minority of clients may likewise be honest-but-curious and may collude, but any coalition below the decryption threshold learns nothing about another

client's displacement, since recovering a plaintext requires the cooperation of the threshold. This is a stronger position than secure-aggregation protocols, where the revealed sum has itself been shown to leak across rounds [49], [50], because in our one-shot protocol the server never obtains a plaintext sum at all.

Because the method fine-tunes a frozen public backbone from a public initialization, there is no alignment phase and *no data-derived quantity leaves a client before encryption*: raw data never leave their client, and the model updates are protected by encryption alone, with no accuracy-reducing perturbation. The one exposure the threat model does not by itself bound is how much an adversary can infer from the final global model, which every party receives in the clear after decryption; we measure that residual exposure directly with membership-inference attacks in section V-I. Robustness to actively malicious or Byzantine clients is outside our scope and we return to it only as a direction for future work.

IV. METHOD

A. Overview

We consider N clients that hold private labelled data over a common label space of C classes and want a shared classifier without pooling their data. Client j holds $\mathcal{D}_j$ of size n_j ; the data are heterogeneous, with each client's class proportions drawn from a Dirichlet distribution of concentration α (smaller α means more skew). Two constraints separate our setting from ordinary federated learning. The protocol is *one-shot*: one upload and one download per client, with no iterative exchange. And privacy is *cryptographic*: a client's contribution is never revealed in the clear, and we are unwilling to buy that privacy with the accuracy loss that differential privacy incurs when it is the only line of defence.

These two constraints largely determine the design. Under encryption the server is blind: it operates on ciphertexts that are by construction indistinguishable from random, so it cannot read a value, compare two values, or branch on the data, and every multiplication it performs spends part of a finite noise budget. The only computation it can afford at scale is a shallow, fixed, linear one — a weighted sum — and a weighted sum of independently trained models is meaningful only when those models already share a frame. A from-scratch federation has no such frame, and building one would take either several rounds or server-side computation on the models themselves, neither of which encryption allows. A *pretrained* model supplies the frame directly: freezing a public backbone fixes the representation every client sees, so the small part each client trains starts from the same place and stays close. Fine-tuning a pretrained model is therefore not a convenience but the learning setting this protocol admits, and it removes the need for any alignment step before training — the property that most distinguishes HE-IFD from prior one-shot federated learning, which spends an extra communication phase, and often a differential-privacy budget, to bring clients into a common frame before they begin.

Concretely, every client uses a frozen public backbone φ followed by a small trainable unit — low-rank adapters together

Algorithm 1 One-shot federated fine-tuning with encrypted aggregation. $\text{Enc}(\cdot)$ denotes encryption under the clients' collective CKKS key.

Require: clients $1, \dots, N$ with data $\mathcal{D}_j$; frozen backbone φ ; public init θ_0 ; steps K

- 1: clients run distributed key generation, yielding a collective CKKS key
- Local fine-tuning — each client j in parallel*
- 2: $\theta_j^{(0)} \leftarrow \theta_0$
- 3: **for** $k = 1$ to K **do**
- 4: $\theta_j^{(k)} \leftarrow \theta_j^{(k-1)} - \eta \nabla \mathcal{L}(f_{\theta}; \mathcal{D}_j)$ ▷ supervised fine-tuning of the trainable unit
- 5: **end for**
- 6: $\Delta_j \leftarrow \theta_j^{(K)} - \theta_0$; upload $\text{Enc}(\Delta_j)$
- Encrypted aggregation — server*
- 7: $\text{Enc}(\theta^*) \leftarrow \theta_0 + \sum_j w_j \cdot \text{Enc}(\Delta_j)$ ▷ PT×CT, CT+CT; depth 1
- 8: broadcast $\text{Enc}(\theta^*)$
- Threshold decryption — clients*
- 9: a threshold of clients jointly decrypt $\text{Enc}(\theta^*)$ to recover θ^*

with a classifier head, with parameters ψ . The backbone never changes, is never transmitted, and is never encrypted; only the trainable unit is fine-tuned, communicated, and operated on under encryption. Keeping that unit small is what makes the encrypted step cheap: it is a few thousand scalars, so a client's entire encrypted contribution fits in one or a few ciphertexts. Every client initializes the trainable unit to the *same* public constant θ_0 , the standard adapter initialization, agreed in advance and carrying no client data. There is no prototype exchange and no peer-to-peer phase; nothing data-derived ever leaves a client before encryption.

The protocol is two steps, shown in fig. 1. Starting from the shared initialization θ_0 , each client fine-tunes its trainable unit on its own data for a bounded number of steps and records the displacement it travelled, $\Delta_j = \theta_j^{(K)} - \theta_0$. The server then combines the encrypted displacements into a single global model, $\theta^* = \theta_0 + \sum_j w_j \Delta_j$, using only homomorphic additions and multiplications by public scalars, after which a threshold of clients jointly decrypts the result.

What we are after is not a model that beats each client on that client's own dominant classes; a local specialist on a narrow slice of the label space is hard to beat on that slice, and we do not claim to. We are after a common model every party can use, that classifies well across the whole label space including classes a given client never observed locally, and that is produced without any party exposing its data or its update in the clear.

B. The Shared Frame

The natural one-shot recipe — let each client train independently and have the server average the results — fails under heterogeneity. When clients begin from different initializations and fit different skewed distributions, their trained parameters occupy different regions of parameter space and their average

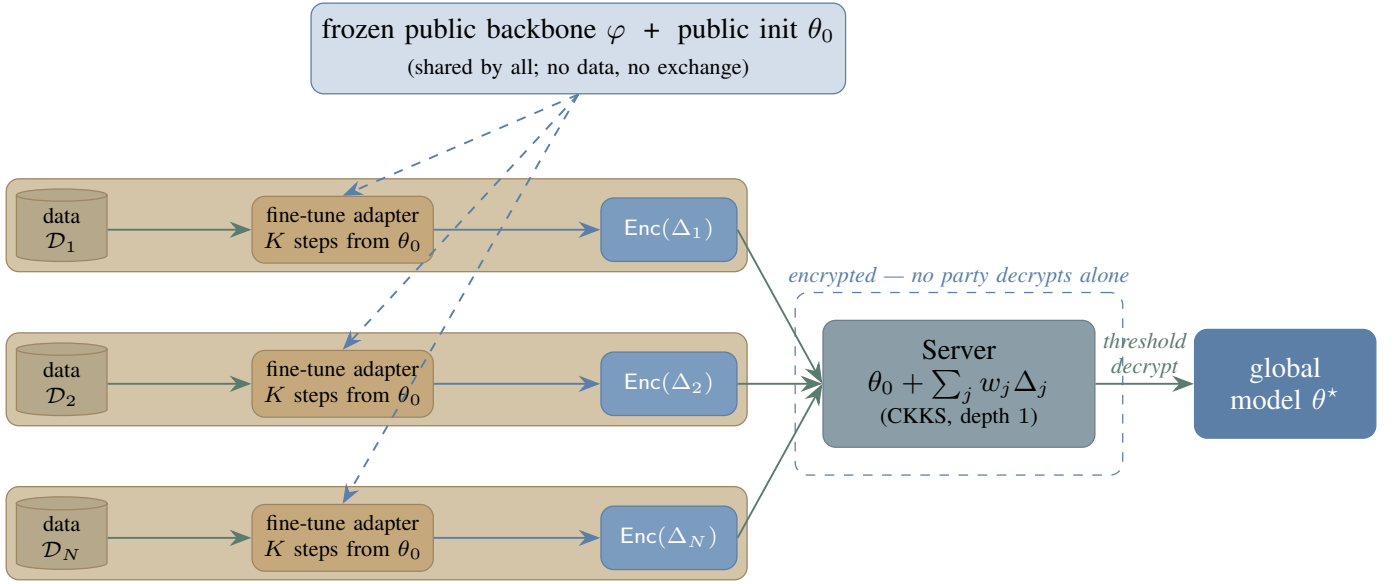


Fig. 1. HE-IFD. A frozen public backbone φ and a public initialization θ_0 are shared by all clients and carry no data, so there is no alignment phase. Each client fine-tunes a small adapter on its own private data $\mathcal{D}_j$ for a bounded number of steps from θ_0 and encrypts the resulting displacement Δ_j . The server's sole operation is the sample-weighted linear combination $\theta^* = \theta_0 + \sum_j w_j \Delta_j$ under multiparty CKKS (additions and public-scalar multiplications only; multiplicative depth one); it never decrypts. A threshold of clients jointly decrypts the global model θ^* .

lies in none of them: the updates point in conflicting directions and cancel when summed. The remedy is for every client to begin from the same point and move only a little, so their displacements live in one frame and add constructively rather than fighting one another.

A pretrained backbone provides this for nothing. Because the representation is fixed and shared, the trainable unit lives in the same parameter space for every client, and initializing it to a common public constant θ_0 places all clients at the same point in that space before any training. No data is consulted to build θ_0 and nothing is exchanged to agree on it; it is a fixed, public initializer. This is exactly the step a from-scratch federation cannot take and must replace with a data-dependent alignment phase — a phase that costs an extra round, exposes a per-client summary, and, when that summary is protected, spends a differential-privacy budget. Fine-tuning a pretrained model removes the phase, and with it the only channel through which client data would otherwise leave in the clear before encryption. We verify in section V that the shared frame is necessary — without it the one-shot combination collapses — and that the frozen backbone supplies it without any alignment phase.

C. Local Fine-Tuning

Starting from θ_0 , client j fine-tunes its trainable unit on its own data for K steps of ordinary supervised training, producing parameters $\theta_j^{(K)}$. The step budget K is deliberately small. This is not an optimisation shortcut but part of the method: a short trajectory keeps the trainable unit close to θ_0 , and therefore inside the shared frame, so the displacement it produces is still expressed in the common coordinates. Allowed to train to convergence, each client would drift to its own local optimum and reintroduce exactly the conflict the shared start was meant to avoid.

At the end of its trajectory the client transmits not its parameters but the displacement

$$\Delta_j = \theta_j^{(K)} - \theta_0, \quad (1)$$

the distance it travelled from the shared start. Because every client departs from the same θ_0 , these displacements are directly comparable across clients, which is what makes the linear combination that follows meaningful. A variant that distills a locally trained teacher into the unit, rather than fine-tuning on hard labels directly, fits the same template and is evaluated as an ablation in section V.

D. Encrypted Linear Aggregation

The clients hold a joint encryption key produced by distributed key generation, so no single party, and in particular not the server, ever holds the secret key. Each client encrypts its displacement Δ_j ; since the trainable unit is a few thousand scalars, this is one or a few ciphertexts. The server fuses the per-client displacements into a single update of the shared model by the sample-weighted combination

$$\theta^* = \theta_0 + \sum_{j=1}^N w_j \Delta_j, \quad w_j = \frac{n_j}{\sum_i n_i}, \quad (2)$$

in which each client contributes in proportion to the data n_j behind it, the weights w_j being public scalars. Forming eq. (2) requires only multiplying each ciphertext by the public scalar w_j and adding the results, with the public θ_0 added as a plaintext; there is no ciphertext-by-ciphertext multiplication and no bootstrapping, so the circuit has multiplicative depth one. A threshold of clients then jointly decrypts θ^* by collective key-switching, and the server never observes a plaintext update. This is the design choice the approach turns on: casting aggregation as an operation that is *linear in the*

encrypted quantities yields a server computation that maps onto the cheapest operations a levelled scheme provides, free of the encrypted nonlinearities that make encrypted training expensive. The operations are also embarrassingly parallel — the clients fine-tune and encrypt independently, and the server’s sum is an associative reduction — so the end-to-end wall-clock is one client’s local computation plus a single aggregation, independent of N . We instantiate eq. (2) in a real multiparty CKKS implementation and report its measured correctness and cost in section V-H.

It is worth being precise about why this works, because eq. (2) can be read too quickly. Since the weights sum to one, the displacement form is algebraically a weighted average of the clients’ fine-tuned units, $\theta^* = \sum_j w_j \theta_j^{(K)}$. That linearity is what keeps the operation cheap to encrypt. But the same average over units trained independently, from different starts, to convergence is the naive combination that fails above. What makes the average land in a good region here is the protocol around it: every $\theta_j^{(K)}$ starts from the shared θ_0 and moves only a bounded distance, so all of them stay in the common frame and their weighted average is itself a sensible model. The contribution is not a new aggregation rule but a one-shot protocol under which a linear, and hence cheaply encryptable, aggregation produces a strong shared model.

E. Security

The model updates are protected cryptographically: the aggregation of eq. (2) runs entirely under multiparty CKKS with threshold decryption, so the displacements Δ_j are never exposed to the server or to any minority of clients, and they are protected without any perturbation. The cryptographic layer adds no accuracy cost. This is the central privacy claim, and the point on which the method departs from differentially private one-shot federated learning, which injects noise into the updates it ultimately releases and so pays a utility tax on the very thing it ships.

We make this precise. Let the view of the honest-but-curious server be everything it receives: the public collective key, the public weights $\{w_j\}$ and initializer θ_0 , the client ciphertexts $\{\text{Enc}(\Delta_j)\}_{j=1}^N$, and the homomorphically computed $\text{Enc}(\theta^*)$. We write $\text{view}_{\text{srv}}(\{\mathcal{D}_j\})$ for this view as a function of the private datasets.

Proposition 1. *Under the IND-CPA security of the multiparty CKKS scheme and its t -out-of- N threshold decryption, for any honest-but-curious server colluding with up to $t - 1$ clients there is a probabilistic polynomial-time simulator $\mathcal{S}$, given only the public parameters, such that $\text{view}_{\text{srv}}(\{\mathcal{D}_j\}) \approx \mathcal{S}$. In particular the server’s view is computationally independent of the private datasets $\{\mathcal{D}_j\}$ and the displacements $\{\Delta_j\}$, and the server cannot recover any plaintext, including θ^* , without a decryption quorum.*

Proof sketch. The simulator outputs the public parameters together with N encryptions of zero under the collective key. By IND-CPA each genuine $\text{Enc}(\Delta_j)$ is computationally indistinguishable from $\text{Enc}(0)$; a hybrid over the N ciphertexts replaces them one at a time, so the whole collection is indistinguishable from the simulated one and carries no information

about $\{\Delta_j\}$ or the data they derive from. The aggregate $\text{Enc}(\theta^*)$ is a deterministic function of those ciphertexts and the public $\{w_j\}, \theta_0$, so it adds nothing. Decryption needs t shares of a secret key that is never reconstructed in one place; the server with at most $t - 1$ clients holds too few shares to decrypt. $\square$

Proposition 1 pins down where information can flow. Because there is no alignment phase, no data-derived quantity leaves a client before encryption, so the protocol exposes private information through exactly one channel, intrinsic to the task rather than an artifact of our construction: the released model itself. After threshold decryption every client holds θ^* in the clear, and white-box access to the jointly built model is unavoidable in any protocol whose purpose is to hand the parties a shared model. The residual information that model carries about training data is what we quantify with membership inference in section V-I.

V. EXPERIMENTS

Our evaluation answers, in order, what the method delivers, why each part of it is needed, what the shared basin contributes, and what privacy and communication cost the cryptographic protocol carries. Throughout, the question we are asking is not whether the global model attains the highest possible test accuracy, but whether the clients can jointly produce a single model that is stronger and more broadly capable than what any of them could obtain alone, and whether they can do so under one round of communication and accuracy-preserving cryptographic privacy.

A. Setup

We evaluate on two regimes that bracket realistic deployments. In the from-scratch regime the feature extractor is a small network trained as part of the protocol: a multilayer perceptron on MNIST and a LeNet-style convolutional network on Fashion-MNIST [51]. In the pretrained regime the feature extractor is a fixed, publicly available backbone whose features are read off without fine-tuning: a ViT-B/32 vision transformer on CIFAR-100 [52], and the RoBERTa-base and MPNet sentence encoders on AG-News. Data are partitioned across clients by a Dirichlet distribution over labels with concentration parameter α (the standard label-skew partition of federated learning: a smaller α leaves each client a few dominant classes and almost none of the rest); we report the heterogeneous ($\alpha = 0.05$), mild ($\alpha = 0.3$), and near-homogeneous ($\alpha = 1.0$) settings. Unless stated otherwise there are $N = 10$ clients, the shared basin is built from raw per-class prototypes, the local distillation runs for a small fixed number of steps K at temperature τ , and every number is averaged over three seeds. Table II summarises the configuration.

We report the test accuracy of the global model, the accuracy of the average and best local teacher as a reference for what a client obtains on its own, the accuracy of a centralized model trained on the pooled data as an upper bound, the accuracy of the shared basin θ_0 on its own, and the coverage of locally-unseen classes, defined as a client’s accuracy on the classes for which it held no local examples.

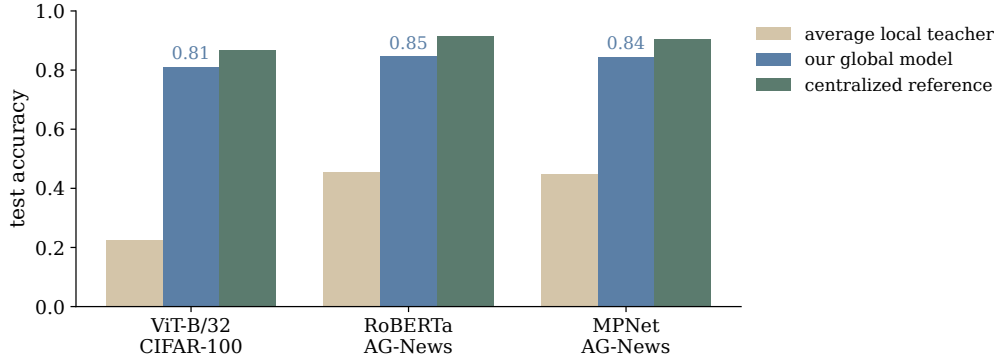


Fig. 2. The protocol turns narrow local teachers into a single, broadly capable model. Under heterogeneous data ($\alpha = 0.05$, $N = 10$), the global model (blue) reaches three to four times the accuracy of the average local teacher and comes within a few points of the centralized reference, on both vision (ViT-B/32, CIFAR-100) and language (RoBERTa, MPNet, AG-News). The frozen backbone never enters the encrypted aggregation, so the protocol and its cost are unchanged across these scales.

TABLE II

EVALUATION DESIGN. EACH AXIS ISOLATES ONE QUESTION RATHER THAN FORMING AN EXHAUSTIVE GRID; RESULTS BELOW VARY ONE AXIS AT A TIME AROUND THE DEFAULT ($N = 10$, RAW PROTOTYPES, THREE SEEDS).

Dimension	Setting	Question it probes
From-scratch vision	MNIST (MLP), Fashion-MNIST (LeNet)	behaviour without a pretrained backbone
Pretrained vision	ViT-B/32 on CIFAR-100	scaling to a strong frozen backbone
Pretrained language	RoBERTa, MPNet on AG-News	the method across modalities
Clients N	5, 10, 20, 50	scalability in participants
Heterogeneity α	0.05 (severe) to 1.0 (mild)	robustness to label skew

B. From Local Specialists to a Shared Global Model

The main experiment shows that the protocol turns a set of narrow local teachers into a single model that is strong across the whole label space. Figure 2 shows this under heterogeneous data ($\alpha = 0.05$) for the pretrained backbones. On each task the average local teacher is weak, because under this partition a client sees only a fraction of the classes and its teacher specialises accordingly: the mean teacher accuracy is 0.23 on CIFAR-100 and below 0.46 on AG-News. The global model produced by the protocol reaches 0.81 on CIFAR-100 and 0.85 on AG-News, between three and four times the average local teacher, and lands within a few points of the centralized reference that was allowed to pool all the data. A client that participates therefore walks away with a model far better than the one it could train alone, and close to what it would have obtained had all the data been centralised, while never having shared that data.

This behaviour is not confined to one backbone scale. The same protocol, with the same bounded distillation and the same linear encrypted aggregation, holds from a small from-scratch MLP up to a frozen vision transformer and a sentence-transformer language encoder. Because the feature extractor is frozen and never enters the encrypted combination, moving to

a larger backbone changes only the quality of the features the small head sees, not the protocol or its cost; the head, and hence the encrypted object, stays the same size. This is what lets the approach scale to modern pretrained models without scaling the cryptographic work, and it is the reason a single classifier head, rather than a fine-tuned backbone, is the right unit to encrypt.

C. Scalability and the Effect of Trajectory Length

Two properties the protocol relies on hold up under direct test. Figure 3 varies the number of clients from five to fifty on ViT-B/32 over CIFAR-100 and shows the global model's accuracy is essentially flat, and if anything improves slightly, as clients are added; more participants do not destabilise the single linear aggregation. Figure 4 varies the length of the distillation trajectory on ResNet-18 over CIFAR-10. With the temperature that keeps the teacher's confident predictions visible, accuracy is highest at a short trajectory and declines gently as the trajectory lengthens, confirming that the bounded K is doing real work: a longer trajectory lets each client drift away from the shared basin and the aggregate degrades toward it. The figure also shows that an over-smoothed teacher signal (a high temperature) erases the gain entirely and worsens with every additional step, which is why we keep both the step budget small and the temperature low.

D. Regularizing the Aggregated Displacement

The trajectory length K is one of two levers that keep clients inside the shared basin; the other is how far the server moves the model along the combined displacement. Writing the aggregate with an explicit scaling coefficient,

$$\theta^*(\lambda) = \theta_0 + \lambda \sum_j w_j \Delta_j,$$

exposes the task-vector scaling coefficient of model merging [53]: λ sets the magnitude of the step away from the basin, and because it is a public scalar the encrypted aggregation stays at multiplicative depth one whatever value it takes. We sweep λ from 0 to 2 by re-scoring the same trajectories, with

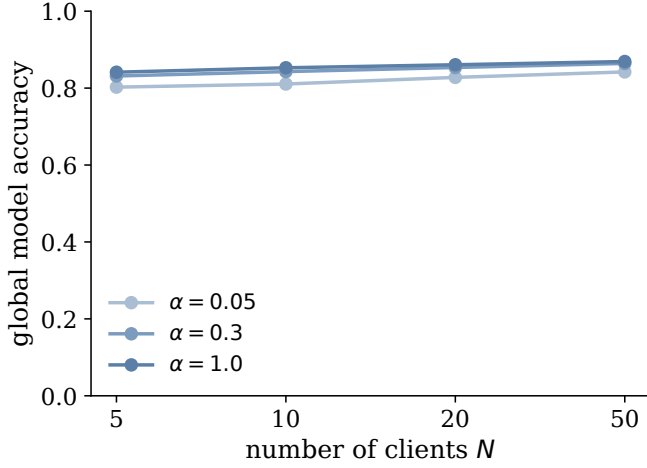


Fig. 3. The global model’s accuracy is stable as the number of clients grows from 5 to 50 (ViT-B/32, CIFAR-100), at every heterogeneity level.

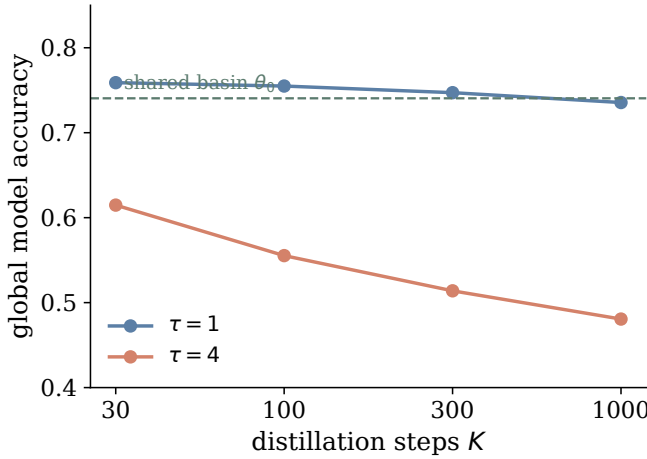


Fig. 4. The bounded trajectory matters. With a low temperature the global model stays just above the shared basin and declines gently as the trajectory lengthens (clients drift away from the basin); a high temperature erases the gain and worsens with every step (ResNet-18, CIFAR-10, $\alpha = 0.05$).

no retraining. The behaviour splits cleanly by heterogeneity. On near-homogeneous data ($\alpha = 1.0$) the optimum is exactly $\lambda = 1$: the per-client displacements already agree, and applying the combined displacement in full is best. Under severe heterogeneity ($\alpha = 0.05$) the optimum moves below 1 and accuracy declines once λ passes 1, the same shape the trajectory-length sweep shows in fig. 4. Both report one mechanism: when clients fit very different local distributions, their displacements applied at full strength carry the aggregate too far out of the common basin, and the combined step has to be regularized. Read this way λ is a drift regularizer, the continuous counterpart of the bounded step budget, and treating it as one rather than pinning it at 1 recovers a few points of accuracy in exactly the heterogeneous regime where drift is worst—on MNIST the heterogeneous optimum adds 2.4 points over $\lambda = 1$ —at no cryptographic cost, with the same downward shift of the optimum holding on the pretrained backbone. The coefficient never settles above 1, which is

the operative statement: under heterogeneity the displacement should be damped toward the basin, never amplified away from it.

E. Comparison with Prior Work

Table III places the method among representative federated-learning approaches along the axes that matter here: whether the protocol is genuinely one-shot, what kind of privacy it provides on the contributions, how many communication rounds it needs, and its principal limitation relative to our goal. Existing one-shot distillation methods are either plaintext, and so offer no privacy on the contributions, or differentially private, and so pay a utility cost on the released model. The homomorphic-encryption methods in federated learning either run for many rounds, as in encrypted training, or are secure-aggregation primitives that sum model updates over an iterative protocol rather than distilling in a single shot. To our knowledge the combination we occupy, a one-shot federated fine-tuning whose aggregation is protected by homomorphic encryption that adds no accuracy cost, has not previously been demonstrated.

Two comparisons are especially informative. Against POSEIDON, the closest cryptographic peer, the accuracy is comparable on MNIST, but POSEIDON reaches it through multi-round encrypted training that the source paper reports at roughly 1.4 hours of wall-clock, whereas our protocol completes in a single round whose server computation is milliseconds and whose communication is a few megabytes. Against the differentially private one-shot methods, the natural utility peers, the contrast is that their accuracy is bought at a privacy budget that perturbs the released model, while our encryption perturbs nothing. The communication advantage is large and structural: a full-model encrypted protocol of the kind the rejected literature assumed would move on the order of hundreds of gigabytes, whereas our single classifier-head ciphertext is a few megabytes per round (section V-H). The protocol is also fully parallel across clients and its aggregation is an associative reduction, so wall-clock does not grow with the number of participants.

[†]FuseFL’s communication equals one-shot but it runs in K block-wise rounds.

[‡]Our $\alpha=0.05$ global-model accuracy on MNIST, CIFAR-100, and AG-News respectively.

F. The Incentive to Participate

It is worth being explicit about what a client gains. The global model does not, in general, beat a client’s own teacher on the handful of classes that client holds in abundance; a specialist trained on a narrow distribution is hard to beat on that distribution, and our aim is not to. What the client gains is coverage of the classes it never saw locally. Under heterogeneous data the global model classifies the locally-unseen classes well above chance, whereas the client’s own teacher cannot classify them at all, having never been trained on them. The value of participating is therefore the breadth that a single party cannot obtain on its own, delivered as a model every party can use, and obtained without exposing any party’s data or update in the clear.

TABLE III

COMPARISON WITH REPRESENTATIVE FEDERATED-LEARNING METHODS. REPORTED ACCURACIES ARE TAKEN VERBATIM FROM EACH SOURCE PAPER AT *its own* SETUP (DATASET, MODEL, N , HETEROGENEITY), SO THE COLUMN IS CONTEXT, NOT A LIKE-FOR-LIKE RANKING; THE PROTOCOL AXES (ONE-SHOT, PRIVACY, ROUNDS, COMMUNICATION) ARE THE CONTROLLED COMPARISON. OUR METHOD IS THE ONLY ONE THAT IS SIMULTANEOUSLY ONE-SHOT, FINE-TUNING-BASED, AND PROTECTED BY ENCRYPTION THAT ADDS NO ACCURACY COST.

Method	One-shot	Privacy	Rounds	Reported acc. (own setup)	Principal limitation vs. our goal
FedDF [18]	no	none	~ 100	rounds-to-target (CIFAR-10)	many rounds; no privacy
FuseFL [21]	no [†]	none	K blocks	84.3% (CIFAR-10)	K rounds; no privacy
DENSE [19]	yes	none	1	data-free (CIFAR-10)	no privacy on contributions
FedLPA [22]	yes	none	1	85.8% (MNIST)	no privacy; needs Fisher info
FedKT [26]	yes	DP-capable	1	90.5% (MNIST)	lossy when private
FedAUXfdp [24]	yes	DP (lossy)	1	75.2% (CIFAR-10, $\epsilon=0.5$)	accuracy falls with budget
FedDiff [25]	yes	DP (lossy)	1	27.8% (CIFAR-10, $\epsilon=10$)	accuracy falls with budget
POSEIDON [4]	no	HE (no noise)	many	89.9% (MNIST, ~ 1.4 h)	multi-round; encrypted training
HE secure agg. [5]	no	HE	per round	protocol only	aggregation, not distillation
HE-IFD (ours)	yes	HE (no noise)	1	84.9 / 81.1 / 84.6%[‡]	—

G. Scope and Limitations

Two boundaries of the approach are worth stating plainly, as both follow from well-understood properties rather than from incidental shortcomings. First, the method aggregates displacements of a head on top of a frozen extractor, so it inherits the quality of that extractor's features: it works best when the frozen representation is already linearly informative for the task. Representations from models trained for next-token prediction are, by their nature, a weaker fit for discriminative classification than those from encoders trained bidirectionally, and we accordingly use bidirectional sentence encoders for the text tasks; the choice of a strong frozen extractor is part of the method, not a tuning detail. Second, the difficulty the shared basin addresses, conflicting client updates under heterogeneity, becomes more severe as the data grow more skewed; this is an inherent property of combining independently trained models in one shot, and the basin is our means of containing it rather than a claim that it disappears. The regime where the basin helps most is precisely the heterogeneous regime where it is most needed, and the experiments above are run in exactly that regime.

H. Cost and Efficiency

The server's only cryptographic operation is the depth-one weighted sum of eq. (2). We implement it end to end in a real multiparty CKKS protocol on the Lattigo library [54], [48] — distributed key generation, encryption under the joint key, the homomorphic aggregation, and threshold decryption by a collective key-switch — and measure what it costs. The cost is set entirely by the number of *fine-tuned* parameters that enter a ciphertext, not by the frozen backbone behind them, so a small adapter and head are cheap whatever model they sit on.

a) Parameters and setup.: All figures use ring degree 2^{14} (8192 slots per ciphertext), a modulus chain $\log Q = \{55, 45\}$ with key-switch prime $\log P = \{61\}$, and scale 2^{45} . The circuit is multiplicative *depth one* — one plaintext-scalar multiplication and a rescale, with no relinearization and no bootstrapping — and decryption is N -out-of- N (no single party can decrypt). Communication figures are exact; timings are single-run wall-clock on one core of a commodity laptop CPU (Apple M4) and are reported as indicative.

TABLE IV

COMMUNICATION PER ROUND (MULTIPARTY CKKS, RING 2^{14}). PER-CLIENT TRAFFIC IS FIXED BY THE TRAINABLE UNIT; TOTALS SCALE LINEARLY IN N . EACH CIPHERTEXT IS 0.5 MiB.

Trainable unit	ct/client	up=down/client	$N=10$	total up=down $N=50$
text head, 4-class (3.1k)	1	0.50 MiB	5.0 MiB	25 MiB
vision head, 100-class (76.9k)	10	5.0 MiB	50 MiB	250 MiB

TABLE V

COMPUTATION TIME (MS, ONE CPU CORE). DKG IS A ONE-TIME SETUP; THE ENCRYPT FIGURE IS PER CLIENT (CLIENTS ENCRYPT IN PARALLEL); AGGREGATION AND DECRYPTION SCALE LINEARLY IN $N \times$ CIPHERTEXTS.

Trainable unit	N	DKG	encrypt/client	aggregate	decrypt
text head, 4-class	10	13	2.3	3.9	2.4
text head, 4-class	100	96	2.2	38.9	23.2
vision head, 100-class	10	11	21.5	39.0	24.6
vision head, 100-class	100	98	21.7	382.2	216.5

b) Communication.: A round is one upload and one download. The per-client traffic is fixed by the trainable-parameter count: a unit of up to 8192 parameters is a single 0.5 MiB ciphertext, and the 100-class vision head (76.9k parameters) spans ten. Total traffic scales linearly in the number of clients (table IV). Even the largest head at $N = 100$ moves about half a gigabyte in total, against the hundreds of gigabytes an encrypted full-model protocol would move.

c) Computation.: The server's homomorphic work is milliseconds at deployment scale and stays sub-second even at $N = 100$ on the largest head (table V). The measured operations follow simple laws: a client encrypts in about 2.2 ms per ciphertext (in parallel across clients), while the server aggregation and the threshold decryption scale as ≈ 0.39 and ≈ 0.23 ms per $N \times$ (ciphertexts/client), and the one-time distributed key generation grows linearly in N at about 1 ms per party. There is no bootstrapping anywhere in the protocol.

d) The adapter budget.: Fine-tuning a low-rank adapter rather than a linear head changes the picture only by the adapter's parameter count. A rank- r adapter on a d -dimensional layer adds $2dr$ parameters, so adapting a handful of layers adds on the order of 10^4 parameters — a small,

fixed number of additional ciphertexts on top of the head. The encrypted object stays a few megabytes per client, the circuit stays depth one, and the server cost grows only with that ciphertext count, never with the size of the frozen backbone the adapter sits in.

e) Correctness.: The decrypted aggregate matches the plaintext computation to a relative ℓ_2 error between 1.4×10^{-9} ($N = 5$) and 2.6×10^{-8} ($N = 100$). This is the encoding precision of CKKS at scale 2^{45} ; it grows mildly with N as ciphertext additions and the key-switch smudging noise accumulate, and it stays several orders of magnitude below any tolerance the decoded model is sensitive to. Because the circuit is depth one, the error is not amplified.

f) Comparison with prior cryptographic federated learning.: The contrast with prior homomorphic federated learning is structural, not incremental. Encrypted-training methods such as POSEIDON [4] reach comparable accuracy but through many rounds of homomorphic computation that the source paper reports at roughly 1.4 hours of wall-clock; secure-aggregation primitives [5] sum updates once per round inside an iterative protocol. Our protocol runs in a single round whose server computation is milliseconds and whose communication is a few megabytes, because the only thing encrypted is a bounded fine-tuning displacement of a small adapter, combined at multiplicative depth one. The broader method-level comparison is in table III.

I. Residual Leakage

The cryptographic guarantee of section IV-E covers the contributions, not the final model, which every party obtains in the clear after decryption. This is not a gap a stronger mechanism could close: any protocol that hands separate parties a shared, useful model must move knowledge between them, and a model that classifies categories a client never saw locally necessarily encodes information about the data that taught it those categories. The meaningful question is not whether residual leakage is zero, which it cannot be, but how large an attack surface a protocol exposes — and our design makes that surface as small as a one-shot federated method can. There is no alignment phase, so nothing data-derived is released before encryption, and the only object exposed in the clear is the single decrypted model.

Read as attack surfaces, the alternatives differ by orders of magnitude. Multi-round federated learning publishes a fresh update every round, so a curious server or peer accumulates a long sequence of views of each client over training [31], [32]. One-shot methods that rely on differential privacy ship, in a single object, essentially the whole of a client's local knowledge — a synthetic copy of its data, or a noised model. HE-IFD exposes only the final aggregate: the per-client displacements are encrypted and never appear in the clear, and because there is no alignment phase there is no prototype channel either.

We measure the residual on that one open channel with membership-inference attacks [33], [34], [35], including the attacks tailored to the federated setting [40], [41], [43], [42], querying the released model both as an external adversary

and as a fellow client that brings its own data as a prior. The leakage stays small: on the language task the strongest shadow-model attack sits at the level of random guessing across heterogeneity levels, so a participant learns essentially nothing about another's membership from the shared model beyond what any usable classifier of that data would reveal.

We do not claim the released model leaks nothing. A model that gives every client coverage of classes it never observed must carry information about the data that supplied that coverage, and an adversary with strong priors about a client's distribution can recover some of it; this is a property of useful shared models, not of our protocol in particular. What the protocol guarantees is that this residual is the *entire* surface — not multiplied across the rounds of an iterative protocol, and not handed over wholesale as released data — and driving the exposed surface down to that inherent floor, rather than promising to remove a leak that cannot be removed when parties set out to build a common model, is the privacy goal the design meets.

VI. CONCLUSION

We presented HE-IFD, a one-shot federated distillation in which the server's only operation on client contributions is a linear aggregation computed under multiparty homomorphic encryption at multiplicative depth one. The construction gives the contributions cryptographic privacy at no accuracy cost, at the communication budget of a single round, and because the feature extractor is frozen and stays outside the encrypted combination, the protocol and its cost are unchanged from small networks to frozen pretrained backbones. The element that makes a one-shot linear aggregation succeed under heterogeneity is the shared loss basin, a cheap and possibly weak common initialization whose purpose is to align the clients' updates rather than to classify well. Across vision and language tasks the method yields a common model far stronger than any client's own and close to a centralized one, a real multiparty CKKS implementation confirms the encrypted aggregation at a few megabytes per round, and the accuracy cost of differentially private basin summaries is negligible across the practical range of privacy budgets. Extending the approach to actively malicious participants, where verifiable homomorphic encryption [55], [56] could certify that the server computed the aggregation honestly, and tightening the analysis of what the released model itself reveals, are natural directions for future work.

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